

List of Formulae

Derivatives

- $(x^n)' = nx^{n-1}$
- $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$
- $(\sqrt{u})' = \frac{u'}{2\sqrt{u}}$
- $(u \cdot v)' = u'v + v'u$
- $\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{v^2}$
- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Rates of Change and Tangent Lines

1. The slope of the tangent line to the graph of $y = f(x)$ at $x = a$, is given by $m = f'(a)$.
2. The equation of the tangent line to the graph of $y = f(x)$ at $x = a$, is given by:

$$y - f(a) = f'(a) \cdot (x - a)$$

If the function $s(t)$ gives the position at time t of an object moving along a straight line, then:

3. The average velocity between the times $t = a$ and $t = b$, is $V_{avg} = \frac{s(b) - s(a)}{b - a}$
4. The instantaneous velocity at $t = a$, is, $v(a) = s'(a)$

Sequences and Series

Arithmetic Sequences and Series	$a_n = a_1 + (n - 1) \cdot d$	$S_n = \frac{n(a_1 + a_n)}{2}$ OR $S_n = \frac{n}{2}[2a_1 + (n - 1)d]$
Geometric Sequences and Series	$a_n = a_1 \cdot r^{n-1}$	$S_n = a_1 \frac{(1 - r^n)}{1 - r}$ OR $S_n = \frac{a_1 - a_n r}{1 - r}$
Infinite Geometric Series	$S = \frac{a_1}{1 - r}$ $ r < 1$	The series S has no Sum $ r \geq 1$

Polar Coordinates

The distance between two polar points $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$, is: $AB = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_1 - \theta_2)}$

The absolute value of the complex number $z = a + bi$ is $|z| = \sqrt{a^2 + b^2}$

Given the complex numbers $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)) \text{ where } z_2 \neq 0$$

If the polar form of a complex number is $z = r(\cos \theta + i \sin \theta)$, then for positive integer n

$$z^n = r^n (\cos n\theta + i \sin n\theta)$$